

# Relazione di Laboratorio - Vulnerability Assessment con Nmap e Metasploitable

## Introduzione

Il laboratorio è stato svolto utilizzando Kali Linux come macchina di analisi e Metasploitable2 come sistema target all'interno di una rete virtuale isolata. L'obiettivo era verificare la raggiungibilità dell'host, individuare le porte aperte, identificare servizi e versioni software e ottenere informazioni dettagliate sul sistema operativo.

## Verifica della connettività

La verifica tramite ping ha confermato che l'host 192.168.60.101 era raggiungibile, con tempi di risposta molto contenuti e nessuna perdita di pacchetti.

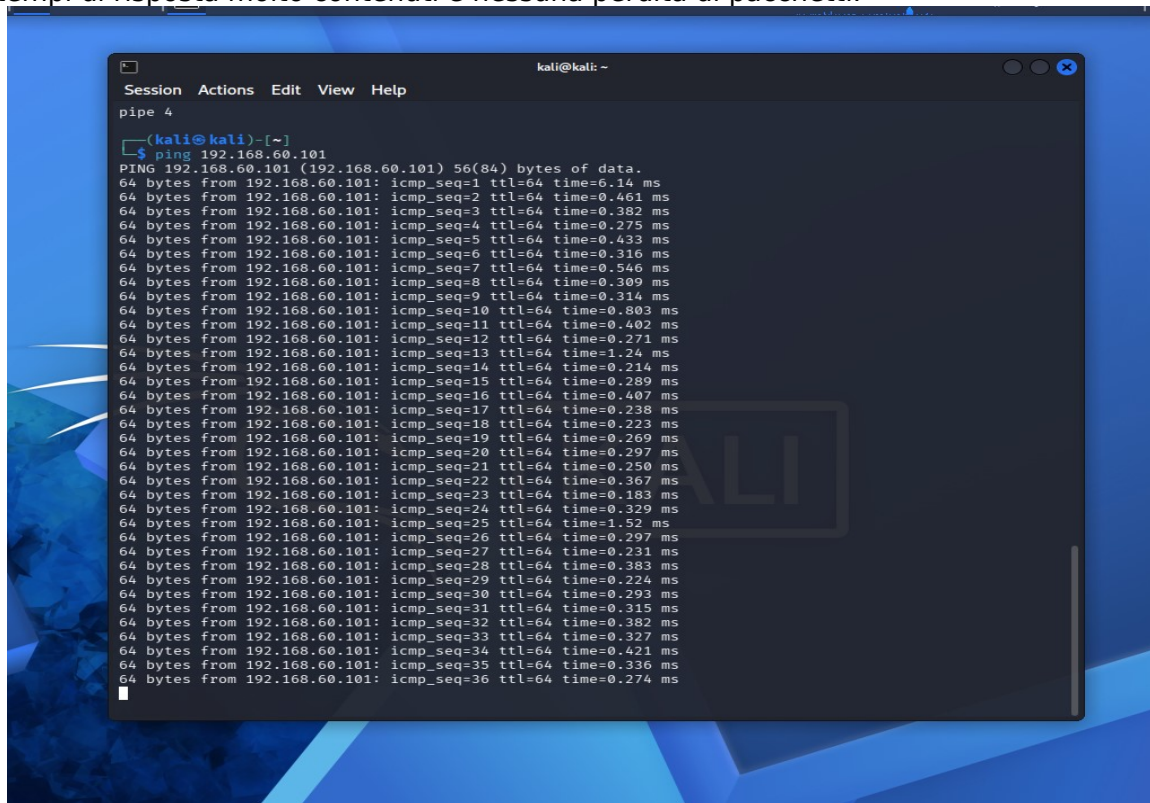
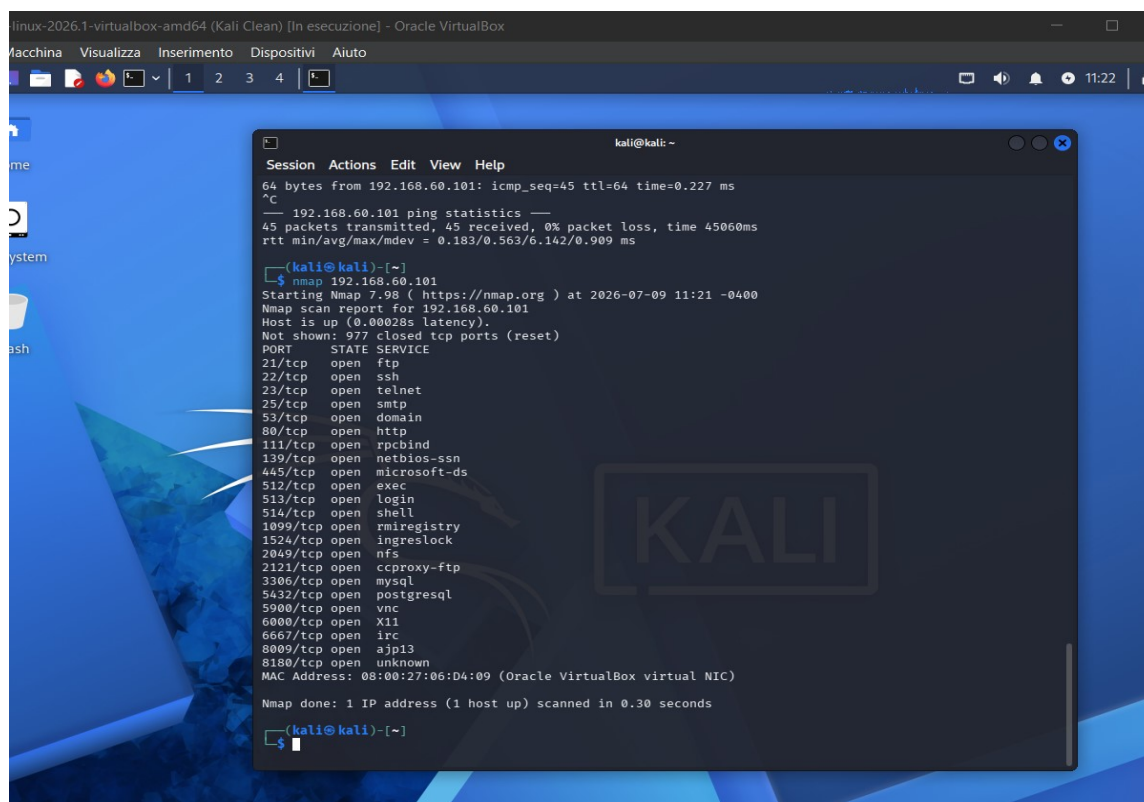
A screenshot of a terminal window titled 'kali@kali: ~'. The terminal shows a 'pipe 4' prompt, followed by a user input '\$ ping 192.168.60.101'. The output shows a successful ping to 192.168.60.101 with 56(84) bytes of data. The output lists 36 ICMP echo requests, each with a sequence number (icmp\_seq=1 to icmp\_seq=36), TTL=64, and a response time in milliseconds. The response times are consistently low, ranging from approximately 0.124 ms to 0.461 ms. The terminal window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'.

Figura 1 - Verifica della connettività tramite ping.

## Scansione base

La scansione standard con Nmap ha evidenziato numerose porte TCP aperte, consentendo una prima mappatura dei servizi esposti dalla macchina vulnerabile.



```
kali@kali: ~  
Session Actions Edit View Help  
64 bytes from 192.168.60.101: icmp_seq=45 ttl=64 time=0.227 ms  
^C  
--- 192.168.60.101 ping statistics ---  
45 packets transmitted, 45 received, 0% packet loss, time 45060ms  
rtt min/avg/max/mdev = 0.183/0.563/6.142/0.909 ms  
  
(kali@kali)~  
$ nmap 192.168.60.101  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-09 11:21 -0400  
Nmap scan report for 192.168.60.101  
Host is up (0.00028s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 08:00:27:06:D4:09 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds  
  
(kali@kali)~
```

Figura 2 - Scansione base con Nmap.

## Service Detection

L'opzione `-sV` ha identificato con precisione i servizi e le rispettive versioni, come Apache 2.2.8, vsftpd 2.3.4, OpenSSH 4.7p1 e PostgreSQL.

```
Kali (Kali Clean) [In esecuzione] - Oracle VirtualBox
mento Dispositivi Aiuto
2 3 4

kali@kali: ~
Session Actions Edit View Help

Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds

kali@kali ~$ nmap -sV 192.168.60.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-09 11:23 -0400
Nmap scan report for 192.168.60.101
Host is up (0.00019s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:06:D4:09 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 52.94 seconds

kali@kali ~$
```

Figura 3 - Identificazione di servizi e versioni.

## Scansione avanzata

Il comando `nmap -A` avvia una scansione avanzata comprendente OS Detection, Version Detection, NSE Script Scanning e Traceroute.

```
(kali@kali) [-]
$ nmap -A 192.168.60.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-09 11:30 -0400
Nmap scan report for 192.168.60.101
Host is up (0.000365 latency)
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_  Connected to 192.168.60.100
|_  Logged in as ftp
|_  TYPE: ASCII
|_  No session bandwidth limit
|_  Session timeout in seconds is 300
|_  Control connection is plain text
|_  Data connections will be plain text
|_  vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_  1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_smtp_commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDST
ATUSCODES, 8BITIME, DSN
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_  bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http_title: Metasploitable2 - Linux
|_http_server_header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|_  program version    port/proto  service
|_  100000 2 111/tcp     rpcbind
|_  100000 2 111/udp     rpcbind
|_  100003 2,3,4 2049/tcp    nfs
|_  100003 2,3,4 2049/udp    nfs
|_  100005 1,2,3 52437/udp   mountd
|_  100005 1,2,2 53662/tcp   mountd
|_  100021 1,3,4 43912/udp   nlockmgr
|_  100021 1,3,4 44236/tcp   nlockmgr
|_  100024 1 37437/tcp   status
|_  100024 1 51663/udp   status
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
|_mysql-info:
|_  Protocol: 10
|_  Version: 5.0.51a-3ubuntu5
|_  Thread ID: 9
|_  Capabilities flags: 43564
|_  Some Capabilities: LongColumnFlag, Support4iAuth, ConnectWithDatabase, SupportsTransactions, Support
sCompression, SwitchToSSLAfterHandshake, Speaks4iProtocolNew
|_  Status: Autocommit
|_  Salt: @n+*.JgJlRc{c[#6`'1q
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
```

Figura 4 - Avvio della scansione avanzata.

## Risultati finali

La parte finale della scansione ha confermato il sistema operativo Linux 2.6.x, la presenza di servizi SMB, informazioni sul traceroute e ulteriori dettagli ottenuti dagli script NSE. Questi risultati completano il processo di ricognizione attiva e forniscono una panoramica tecnica completa del target.

```

513/tcp open  login?
514/tcp open  shell          Netkit rshd
1899/tcp open  java-rmi        GNU Classpath grmiregistry
1924/tcp open  bindshell      Metasploitable root shell
2049/tcp open  nfs            2-4 (RPC #100003)
2121/tcp open  ftp            ProFTPD 1.3.1
3306/tcp open  mysql          MySQL 5.0.51a-3ubuntu5
|_ mysql-info:
|_ Protocol: 10
|_ Version: 5.0.51a-3ubuntu5
|_ Thread ID: 9
|_ Capabilities flags: 43564
|_ Some Capabilities: LongColumnFlag, Support4iAuth, ConnectWithDatabase, SupportsTransactions, Support
sCompression, SwitchToSSLAfterHandshake, Speaks4iProtocolNew
|_ Status: Autocommit
|_ Salt: @n+*.JgJlRc[c[#6`1q
5432/tcp open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
|_ ssl-cert: Subject: commonName=ubuntu04-base.localdomain/organizationName=OCOSA/stateOrProvinceName=Th
ere is no such thing outside US/countryName=XX
|_ Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-16T14:07:45
|_ ssl-date: 2026-07-09T15:32:27+00:00; +1s from scanner time.
5900/tcp open  vnc            VNC (protocol 3.3)
|_ vnc-info:
|_ Protocol version: 3.3
|_ Security types:
|_ VNC Authentication (2)
6600/tcp open  x11            (access denied)
6667/tcp open  irc            UnrealIRCd
8009/tcp open  ajp13         Apache Jserv (Protocol v1.3)
|_ ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http          Apache Tomcat/Coyote JSP engine 1.1
|_ http-favicon: Apache Tomcat
|_ http-title: Apache Tomcat/5.5
MAC Address: 08:00:27:06:D4:09 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:
linux:linux_kernel

Host script results:
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_ smb-os-discovery:
|_ OS: Unix (Samba 3.0.20-Debian)
|_ Computer name: metasploitable
|_ NetBIOS computer name:
|_ Domain name: localdomain
|_ FQDN: metasploitable.localdomain
|_ System time: 2026-07-09T11:31:06-04:00
|_ lock-skew: mean: 1h20m01s, deviation: 2h18m34s, median: 0s
|_ smb-security-mode:
|_ authentication_level: user
|_ challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_ smb2-time: Protocol negotiation failed (SMB2)

TRACEROUTE
HOP RTT ADDRESS
1 0.36 ms 192.168.60.101

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 151.52 seconds

kali@kali:~$

```

Figura 5 - Risultati finali della scansione avanzata.

## Conclusioni

L'attività ha dimostrato l'utilità di Nmap nelle fasi di Reconnaissance attiva e Vulnerability Assessment, permettendo di raccogliere informazioni fondamentali su host, porte, servizi, versioni software e sistema operativo prima di eventuali attività di analisi delle vulnerabilità.